Chapter 1: Coupled Simulation

In this chapter, we couple the locally mass-conserving MFEM and FVM methods to perform numerical simulations of a partially molten mantle. We describe the implementation details and coupling strategy that integrate the MFEM, FVM, and the eutectic phase package, with the remaining uncertainty in porosity ϕ addressed through phase package evaluation. Before tackling the fully coupled problem, we first examine an alternative porosity evolution scenario based on a prescribed melting rate. We then present an example that combines the MFEM, FVM, and eutectic phase package into a fully coupled computation. The discussion begins with pseudo-1D scenarios and subsequently extends to full 2D simulations.

1.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ denote the cell-averaged solution to the transport system (**??**) at time t^n. Similarly, Let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ represent the solution to the Darcy-Stokes system at the same time step t^n. Finally, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n obtained from the transport result $\mathcal{U}^n$.

For each time step t^n, we begin by reconstructing the dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ using ML-WENO reconstruction at each data points. The eutectic phase package then provides the corresponding phase parameters $\mathcal{W}^n$ based on these point wise values. With $\mathcal{W}^n$ determined, we solve the Darcy-Stokes system at the fixed time t^n to obtain $\mathcal{V}^n$. Finally, the cell-averaged solution $\mathcal{U}^{n+1}$ at the next time step t^{n+1} is then obtained by forward marching of the transport system. The complete procedure is summarized as an coupled solver in algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let U^0 be given as initial data.

1: Initialize $\mathcal{U}^0$, compute $\mathcal{W}^0$ with eutectic phase package.

2: Compute ϕ^0 in $\mathcal{W}^0$, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^0$.

3: **for** Time level $n = 1, 2, \dots, n_{\max} - 1$ **do**

4: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.

5: Compute ϕ^n in W^n, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^{n+1}$.

6: Given $\mathcal{W}^n$ and $\mathcal{V}^{n+1}$, compute dimensionless effective velocity (**??**) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi\check{\mathbf{v}}_l + D_i(1-\phi)\check{\mathbf{v}}_s}{\phi + (1-\phi)D_i}, \quad \check{\bar{\mathbf{v}}} = \phi\check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

7: Explicitly marching forward the transport system (**??**) to next time $n+1$ for $\mathcal{U}^{n+1}$.

8: **end for**

1.1.1 Discontinuous Porosity

Due to the presence of no melt regions and distinct phase regions, the reconstructed values of the dimensionless enthalpy and mass concentration may differ on the two sides of an edge shared by adjacent elements. This, in turn, can lead to two different porosity values being computed for the same edge. However, in a conforming MFEM scheme, it is essential to assign a single consistent and physically appropriate porosity value on each element edge.

Therefore, we evaluate the geometric average of the two values of porosity ϕ_e^+ and ϕ_e^- at each quadrature points on both sides of an element edge e as

$$\overline{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}, \tag{1.1}$$

where ϕ_e^+ and ϕ_e^- are porosity evaluated from two sides of the element edge e. The geometric averaged porosity $\overline{\phi}_e$ is then used in the continuous compaction equation (**??**) , where the face integral is converted to an edge integral via the divergence theorem.

Importantly, when the porosity on one side of the edge vanishes, the geometric average also becomes zero. This property enforces the physical condition that no liquid flow can pass through an edge adjacent to a no-melt region.

1.2 Pseudo One-Dimensional Column Tests

In this section, we test the coupled algorithm on a two-dimensional $M \times N$ grid with a pseudo one-dimensional compaction column problem. We study a compacting column aligned along the y-direction, defined over the domain $y \in [-H, 0]$ and $x \in [-L, L]$. Two scenarios are investigated:

- Prescribed melting case: The MFEM and FVM solvers are coupled without phase computation. Instead, a simple prescribed melting function $\Gamma(\mathbf{x})$ is used;
- Phase calculation case: Phase transition and porosity evolution are computed simultaneously, providing a complete coupling between the MFEM, FVM, and eutectic phase package.

For the two tests presented below, we take domain dimensions $H = 2$ and $L = 0.1$. We fix mesh resolution in x-direction as $M = 2$ and pick different resolutions in y-direction. We fix the parameter $\Theta = 0$ as well. The code is implemented in PETSc/C++, with implementation details attached as appendix.

1.2.1 Prescribed Melting Cases

Before coupling mechanical, transport and phase package altogether into one integrated computation, we first illustrate a simpler model that couples MFEM and FVM solvers without involving the phase package. In this reduced setting, the porosity distribution is not determined by phase-splitting calculations. Instead, we prescribe the melting rate or equivalently, the porosity evolution through a fixed source function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{1.2}$$

We express conservation of liquid mass by introducing the prescribed melting term $\Gamma(\mathbf{x})$ as a source term on the right-hand side of the advection equation as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \tag{1.3}$$

where the diffusion term is being dropped for simplicity in this case.

We prescribe the following boundary conditions for the MFEM and the FVM solvers respectively to replicate the behavior of the one-dimensional problem within the two-dimensional grid.

For the Darcy-Stokes MFEM solver:

- Bottom Edge $y = -H$;
  - Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions with a constant upwelling velocity v:

    $$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e v\, ds.$$

  - Tangential Stokes velocity is prescribed with essential boundary condition:

    $$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0.$$

  - Normal Darcy flux is prescribed with essential boundary condition:

    $$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

- Top Edge $y = 0$;
  - Normal Stokes velocity and supplemental edge averaged flux are prescribed with natural boundary conditions: traction free

    $$t_0 = 0;$$

- Tangential Stokes velocity is prescribed with zero essential boundary condition:

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0.$$

- Normal Darcy flux is prescribed with natural boundary condition:

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

- Left and Right Edges $x = \pm L$;
  - Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions:

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0, \; |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} \, ds = 0;$$

  - Tangential Stokes velocity is prescribed with the natural boundary condition: free traction

$$t_0 = 0;$$

  - Normal Darcy flux is prescribed with non-permeable essential boundary condition:

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

For the transport FVM solver:

- Bottom Edge $y = -H$;
  - A corresponding Dirichlet boundary condition is prescribed on the inflow boundary aligning with the essential boundary condition specified in the MFEM solver;
- Top Edge $y = 0$;
  - A free outflow boundary condition is imposed, consistent with the free traction boundary condition applied in the MFEM solver;

- Left and Right Edges $x = \pm L$;
  - A no-flow boundary condition is prescribed in consistence with the zero-flow Dirichlet boundary used in the MFEM solver.

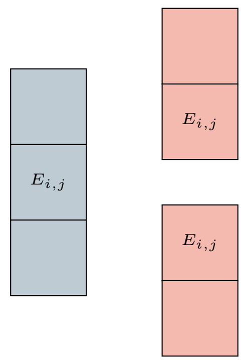

Figure 1.1: An exhibition of a set of 1D ML-WENO (3,2) stencils on a rectagular mesh.

For simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 1.1, since higher-order accuracy in the xx-direction is unnecessary for this problem. We use second order SSP Runge-Kutta time integration scheme defined in **??**.

We consider two different scenarios, each characterized by distinct prescribed melting functions and initial conditions:

Case 1: A piece-wise constant source function.

$$\Gamma_1(z) = \begin{cases} 2.5e-3, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \tag{1.4}$$

We assign an initial condition of no melting everywhere as

$$\phi_1(z,0) = 0.0, \quad z \in [-2.0, 0.0], \tag{1.5}$$

Figure 1.2 illustrates the temporal evolution of porosity, Stokes and Darcy velocities, Stokes and Darcy pressures, as well as the effective pressure, defined as

$p_{\text{eff}} = p_s - p_l$. The third panel, corresponding to dimensionless time $\check{t} = 1325$, demonstrates that the system has reached a steady state after sufficient evolution.

Case 2: We next consider a slightly more complicated scenario in which a melting source is trapped beneath the bottom of an existing porosity channel. The trapped melting source produces a step in porosity, thereby induces an instantaneous perturbation in the compaction rate.

$$\Gamma_2(z) = \begin{cases} 0, & -1.5 < z \leq 0, \\ 1.5e-3, & -1.8 < z \leq -1.5, \\ 0, & -2.0 \leq z \leq -1.8, \end{cases} \tag{1.6}$$

where we assign an initial condition with an existing porosity channel as

$$\phi_2(z,0) = \begin{cases} 5e-2, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \tag{1.7}$$

Figure 1.3, 1.4 and 1.5 illustrate the temporal evolution of porosity, Stokes and Darcy velocities and Stokes and Darcy pressures. A propagating discontinuity in porosity forms when the porosity profile decreased upward Katz (2022). Eventually, the system reaches a steady state, characterized by a narrower porosity channel. The simulation demonstrates robust and stable evolution, despite the presence of a highly complicated porosity profile.

1.2.2 Phase Coupled Case

In this section, we couple the simulation with the eutectic phase package calculation. In this scenario, we transport nondimensional bulk composition and enthalpy instead of transporting porosity directly, but compute porosity from the transport variables.

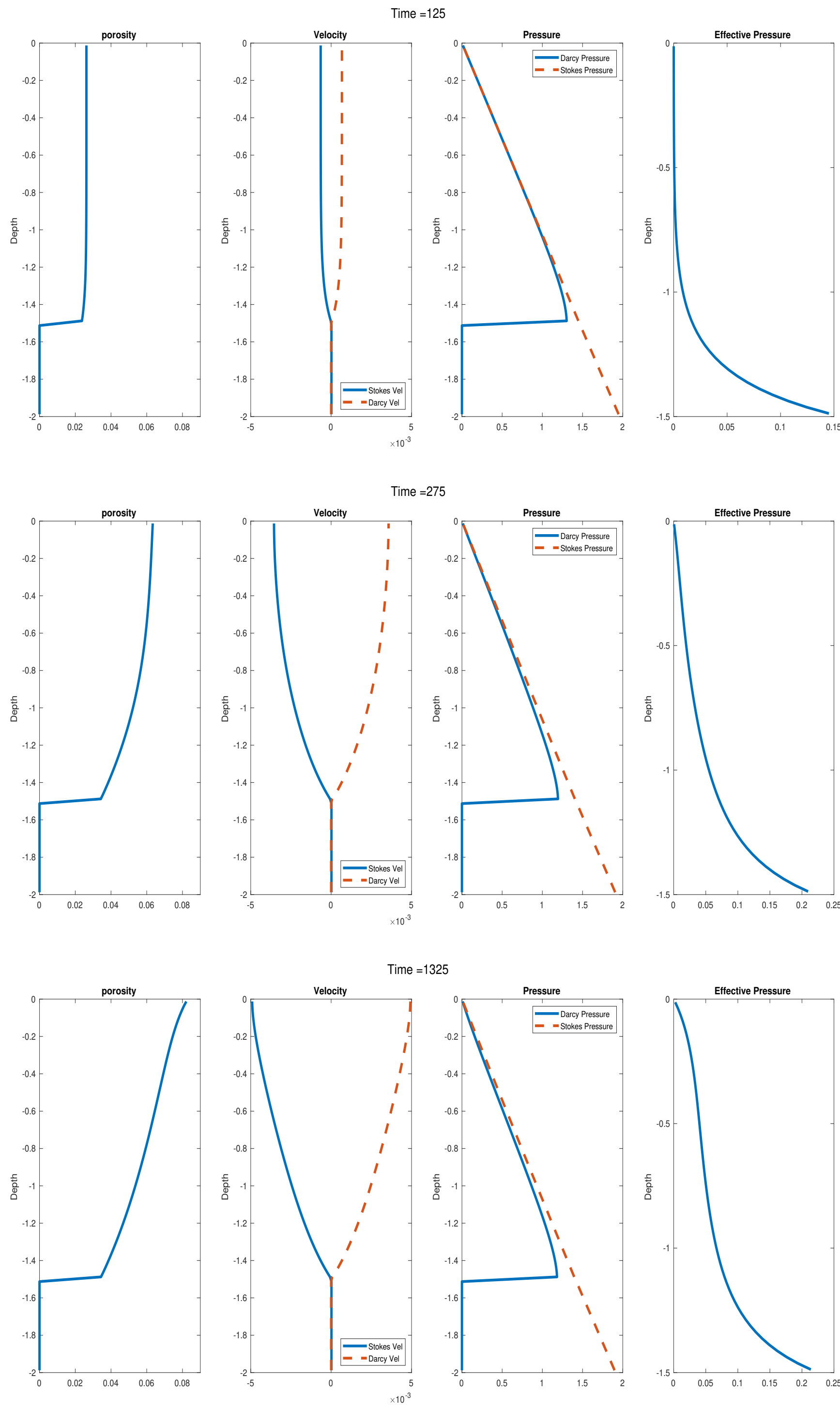

Figure 1.2: Prescribed melting case with constant melting source term 1.4. Shown at time = 125, 275 and 1325, where the system reaches steady state at the time = 1325.

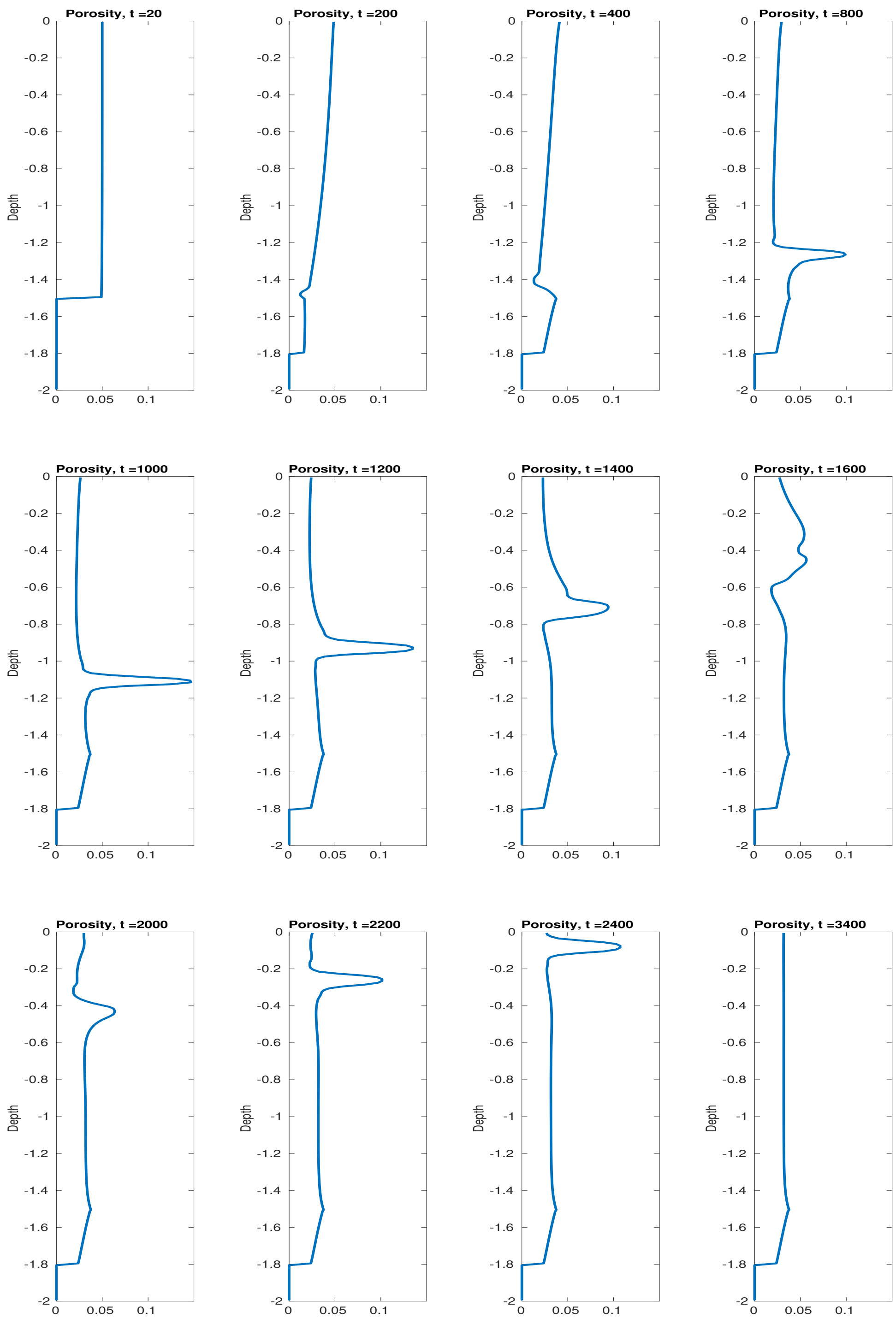

Figure 1.3: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of porosity shown at different time steps.

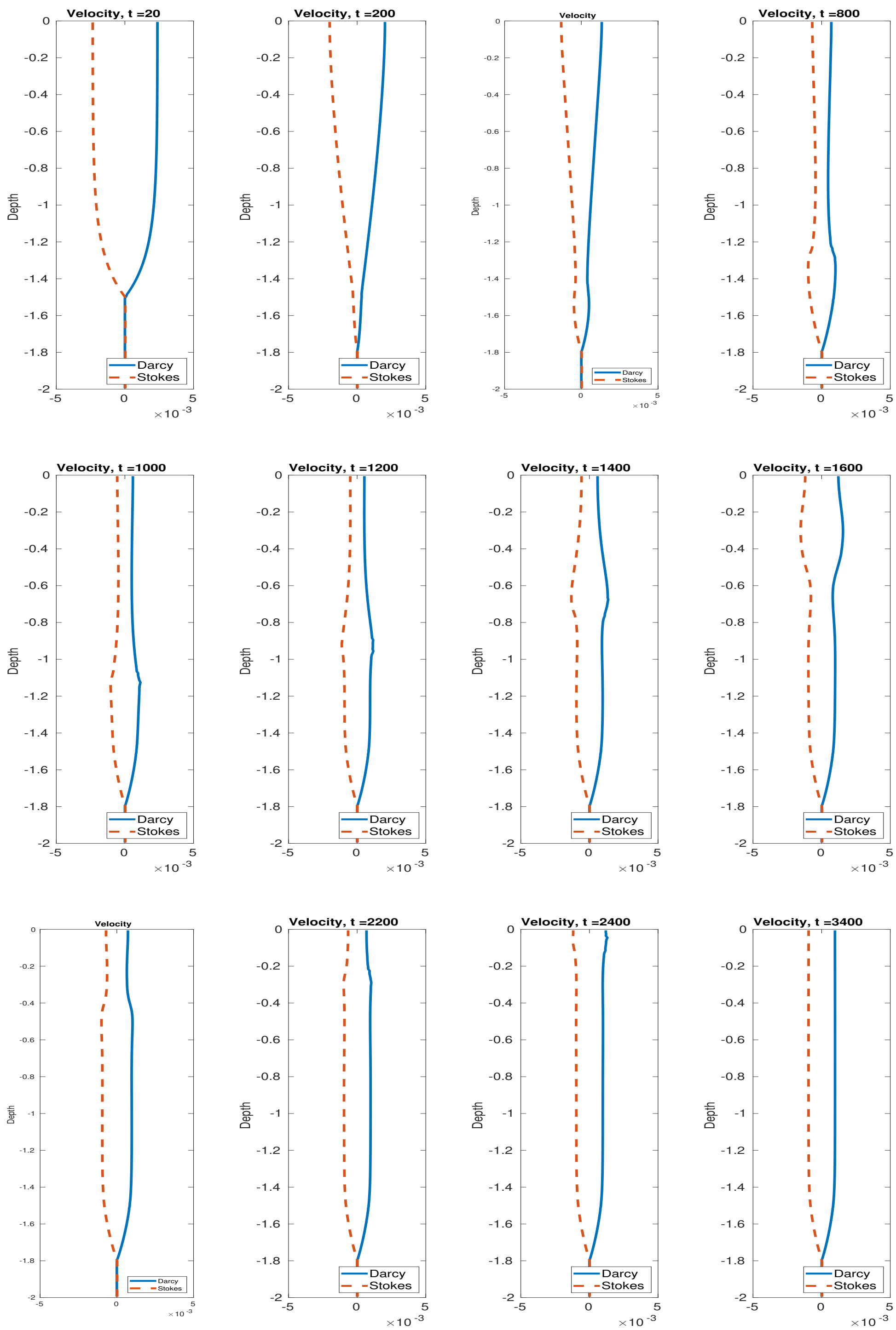

Figure 1.4: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of velocities shown at different time steps.

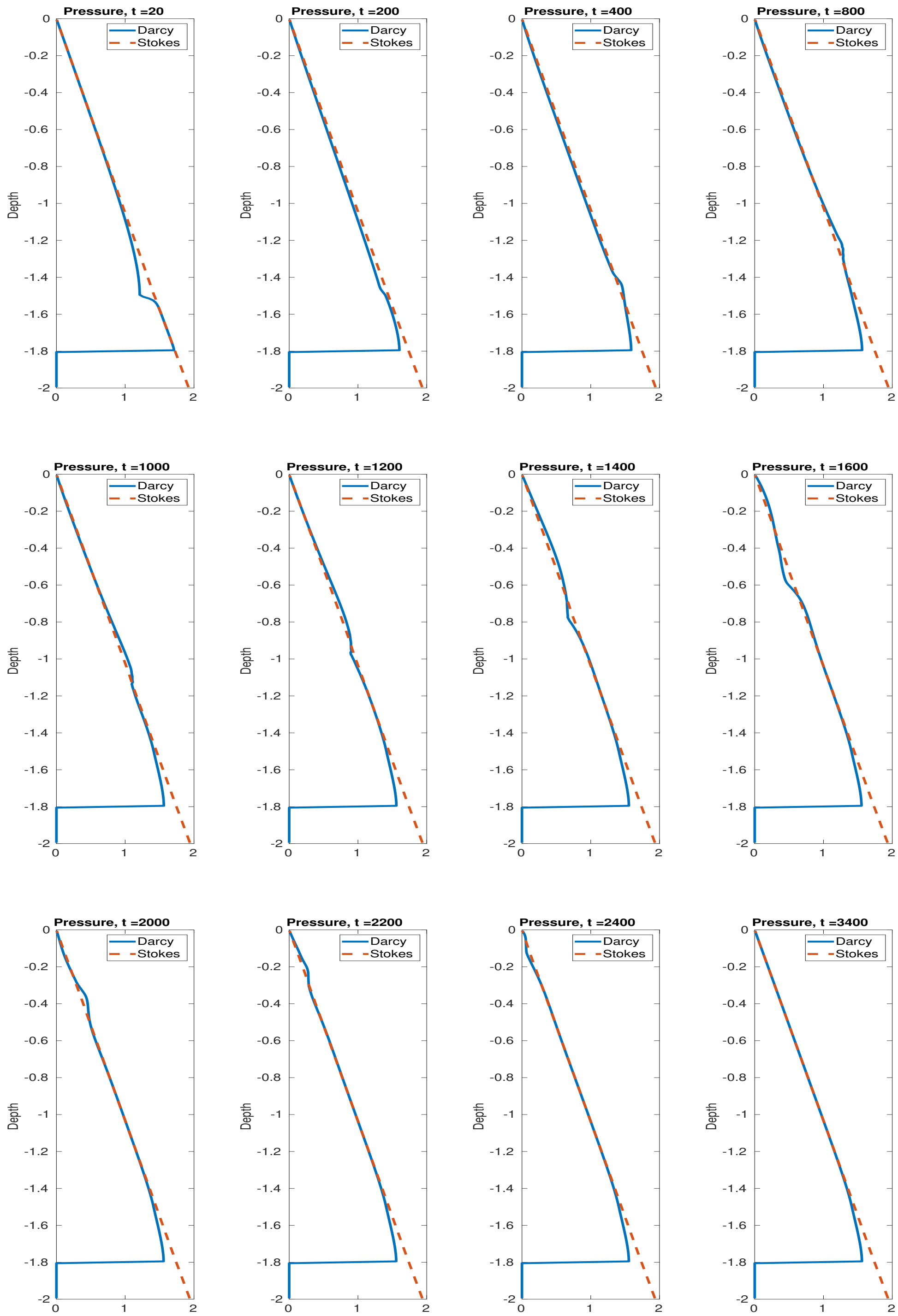

Figure 1.5: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of pressures shown at different time steps.

Works Cited

Richard F. Katz. *The Dynamics of Partially Molten Rock*. Princeton University Press, 1st edition, 2022. ISBN 9780691232645.